

**Practicum No.3****Date: 13/02/2025****Business Idea Generation for CSE/AIML/DSCS/DS Venture****I. Practical Significance**

This practicum introduces students to the creative and analytical process of generating innovative business ideas in the domains of Computer Science, Artificial Intelligence & Machine Learning, Data Science & Cyber Security, and Data Systems. The objective is to cultivate an entrepreneurial mindset by encouraging students to identify real-world problems in society, industry, and digital ecosystems and translate them into feasible, technology-driven business opportunities.

Through this exercise, students apply structured idea-generation techniques such as brainstorming, SCAMPER, and mind mapping to develop novel startup concepts—including AI-powered healthcare assistants, intelligent cybersecurity platforms, data-driven decision support tools, cloud-based automation services, and personalized learning systems. By completing this practicum, students enhance their creativity, problem-solving abilities, and entrepreneurial awareness, forming a strong foundation for innovation-oriented careers in the computing and AI sectors.

**II. Competency and Industry-Specific Practical Skills**

This practicum exercise is expected to develop the following skills for the industry-identified competency:

- **Develop** a Pitch deck, project proposals for small/medium-scale startups logically and scientifically.
- a) **Conceptualize** high-impact CSE product opportunities in AI-driven platforms, autonomous systems, digital twins, and next-generation software solutions.(CDO).
- b) **Use** technically robust survey tools for software engineering, cloud-native systems, and intelligent computing sectors.(CDO).
- c) **Interpret** target customer clusters for SaaS applications, AI-powered automation tools, and data-driven service platforms.(CDO).
- d) **Design** sustainable business architectures for scalable digital, software, and data-centric ventures.(CDO).

**III. Relevant CO(s)**

- **CO1. Identify** entrepreneurial traits and match them with suitable business opportunities.

**IV. Practical Outcome (PrO)**

- **PrO3 – Generate** a list of startup ideas in electronics/communication/electrical/biomedical domains (e.g., smart wearables, affordable hearing aids, smart energy audit system).

**V. Related ADO(s)**

- a) **Exhibit** innovative and analytical thinking.
- b) **Communicate** ideas effectively in teams.
- c) **Demonstrate** ethical and sustainable decision-making while proposing ideas.

**VI. Minimum Underpinning Theory**

Business Idea Generation is the process of identifying, developing, and refining new business opportunities based on existing problems, emerging technologies, and market needs. In engineering entrepreneurship, it combines technical knowledge, market understanding, and creative thinking to propose feasible product or service ideas.

Common methods used include:

1. **Brainstorming** – Generating multiple ideas collaboratively without judgment.
2. **SCAMPER Technique** – Modifying existing products using: *Substitute, Combine, Adapt, Modify, Put to another use, Eliminate, and Rearrange*

3. **Mind Mapping** – Visualizing relationships between problems, technologies, and possible solutions
4. **Problem-Based Ideation** – Identifying pain points in real-life scenarios and finding engineering solutions

For example:

- In CSE, a viable startup idea could be a cloud-native intelligent computing platform capable of handling scalable data processing, AI model deployment, real-time analytics, and secure service delivery for diverse industry applications.
- In AIML, it could be an AI-powered medical diagnosis assistant.
- In DSCS, it might be a threat intelligence platform using machine learning for cyberattack prediction.
- In DS, a strong idea could be a data-driven decision-support dashboard for SMEs.

The aim is to nurture **entrepreneurial thinking** — identifying viable opportunities and developing a portfolio of innovative ideas that can evolve into real ventures.

## VII. Practicum Set-up/Circuit Diagram/Algorithm/Flowchart

No special lab setup required; group or individual activity in class/computer lab

## VIII. Resources Required

| S. No. | Name of Resource                      | Suggested Broad Specification       | Quantity    |
|--------|---------------------------------------|-------------------------------------|-------------|
| 1      | Laptop/PC                             | For research and documentation      | 1 per team  |
| 2      | Internet access                       | For identifying trends and problems | As required |
| 3      | Brainstorming worksheet / chart paper | For idea listing and mind mapping   | 1 set       |
| 4      | Reference materials                   | Magazines, journals, websites       | 1 per team  |

## IX. Safety Precautions

- a) **Ensure** ideas are original and not plagiarized.
- b) **Avoid** using confidential or proprietary data.
- c) **Follow** ethical and social boundaries while developing ideas.

## X. Procedure

1. **Form** a team and select a domain of interest.
2. **Discuss** current problems or opportunities in that domain.
3. **Conduct** a brainstorming or SCAMPER session.
4. **List** all ideas generated during the session.
5. **Filter** out impractical or repetitive ideas.
6. **Refine** the shortlisted ideas with more details (target users, technology, benefit).
7. **Present** your findings in a report or slide deck.

**XI. Actual Resources Used**

| S. No. | Name of Resource        | Broad Specifications   |                                                         | Quantity | Remarks (If any)           |
|--------|-------------------------|------------------------|---------------------------------------------------------|----------|----------------------------|
|        |                         | Make                   | Details                                                 |          |                            |
| 1.     | Laptop / PC             | ASUS                   | Intel i5 / Ryzen 5 Processor, 8GB RAM                   | 1        | research and documentation |
| 2.     | Miro Tool               | Miro (Online Platform) | Free online brainstorming template for SCAMPER          | 1        | Organized ideas visually   |
| 3.     | YouTube Learning Videos | YouTube Platform       | SCAMPER technique tutorial & Visual Brainstorming video | 1        | Concept understanding      |

**XII. Actual Procedure Followed**

1. A team of four members was formed and the domain **AIML (Artificial Intelligence & Machine Learning)** was selected.
2. The team discussed various real-world problems in healthcare, education, cybersecurity, and student life.
3. After discussion, the problem of **early stress detection and mental health support for students** was identified.
4. Each SCAMPER component (Substitute, Combine, Adapt, Modify, put to another use, Eliminate, Reverse) was applied systematically to the identified problem.
5. All ideas were written without filtering during the brainstorming stage.
6. The ideas were evaluated based on:
  - Technical feasibility
  - Market demand
  - Scalability
  - Innovation potential
7. The most viable idea, **Mind Guard AI – Preventive AI Wellness Coach**, was selected.

**XIII. Observations and Data**

| S. No. | Entrepreneur             | Major Venture              | Key Traits           | Opportunity Identified |
|--------|--------------------------|----------------------------|----------------------|------------------------|
| 1      | AI Mental Health Chatbot | 24/7 support               | Medium               | High                   |
| 2      | Smart Stress Monitoring  | Wearable integration       | Medium               | Medium                 |
| 3      | AI Mood Journal          | Sentiment analysis         | High                 | High                   |
| 4      | Burnout Predictor        | Academic stress prediction | High                 | High                   |
| 5      | Emotion Voice Detection  | Real-time emotion scan     | Low (privacy issues) | Medium                 |

**XIV. Results/Observations**

1. The SCAMPER method significantly increased idea diversity.
2. Combining different perspectives resulted in more innovative solutions.

3. Preventive AI solutions showed stronger long-term scalability.
4. Privacy and ethical considerations played an important role in filtering ideas.
5. Structured evaluation helped in selecting the most viable idea scientifically.

## XV. Interpretation of Results

The structured ideation approach proved more effective than random brainstorming.

- Transform a simple problem into multiple innovative solutions.
- Explore both incremental improvements and disruptive ideas.
- Evaluate feasibility logically instead of emotionally.

## XVI. Conclusions.

The practicum successfully demonstrated how structured idea-generation techniques like SCAMPER can lead to innovative and scalable startup ideas in the AIML domain.

- High technical feasibility
- Strong market demand
- Scalable SaaS model
- Positive social impact

## XVII. Suggested Practicum Related Questions *(Every batch will be given another set of questions)*

1. **Analyze** how emerging technology trends in CSE, AIML, cybersecurity, and data science can be mapped to unsolved societal or industry problems to generate viable business ideas.
2. **Evaluate** the effectiveness of structured ideation techniques (e.g., SCAMPER, Design Thinking, TRIZ) in generating innovative product or service ideas for technology-driven ventures.
3. **Identify** a real-world problem in healthcare, finance, agriculture, or smart cities and propose a technology-enabled solution using CSE/AIML/DSCS/DS capabilities.
4. **Assess** the feasibility of proposed startup ideas by examining technical complexity, market demand, scalability, and competitive landscape.
5. **Examine** how data availability, API ecosystems, and cloud infrastructure can influence the scope and architectural design of new tech-based business ideas.
6. **Evaluate** the role of AI/ML model maturity, dataset requirements, and computational constraints in shaping the viability of AI-driven business solutions.
7. **Analyze** how cybersecurity considerations such as data privacy, threat models, and regulatory compliance impact the design of a proposed startup idea.
8. **Compare** multiple business ideas generated during the practicum and justify the selection of the most viable one based on innovation potential, customer value, and revenue model.
9. **Investigate** how user personas and customer journey mapping can support the formulation of high-impact digital products in CSE, AIML, DSCS, and Data Science domains.
10. **Design** a problem-solution fit analysis for an IoT, cloud-native, blockchain, or data-driven startup idea and justify its relevance to current industry needs.

## XVIII. References / Suggestions for further reading

Timmons & Spinelli, New Venture Creation: Entrepreneurship for the 21st Century, McGraw-Hill, 2019

## XIX. Suggested Assessment Scheme

The given performance indicators should serve as a guideline for assessment regarding process and product-related marks.

**Note:** The weightage for the process-related skills and product-related skills will change depending on the PrO, COs, and the related competency of the concerned course.

| Performance Indicators                        |                                                  | Maximum Marks | Marks Obtained |
|-----------------------------------------------|--------------------------------------------------|---------------|----------------|
| <b>Process-related: 18 Marks - 60%</b>        |                                                  |               |                |
| 1                                             | Identification of focus domain and problem areas | 3             |                |
| 2                                             | Use of structured idea generation techniques     | 4             |                |
| 3                                             | Team collaboration and creativity                | 4             |                |
| 4                                             | Feasibility evaluation and reasoning             | 3             |                |
| 5                                             | Ethical originality and clarity of ideas         | 4             |                |
| <b>Product-related Skills: 12 Marks - 40%</b> |                                                  |               |                |
| 1                                             | Quality and novelty of generated ideas           | 3             |                |
| 2                                             | Logical presentation and organization            | 3             |                |
| 3                                             | Clarity of findings and conclusions              | 3             |                |
| 4                                             | Report/presentation submission                   | 3             |                |
| 5                                             | Innovation potential and market relevance        | 3             |                |
| <b>Total</b>                                  |                                                  | <b>30</b>     |                |

*To be filled by Student Team Leader:*

| S.No. | Name of the Student | Register No. |
|-------|---------------------|--------------|
| 1     | NEAVIS RISHVA. J    | URK25CS1236  |

*To be filled by the Faculty Member:*

| Marks Obtained              |                             |            | Name and Designation of the Faculty Member | Signature with Date |
|-----------------------------|-----------------------------|------------|--------------------------------------------|---------------------|
| Process-Related skills (18) | Product-Related skills (12) | Total (30) |                                            |                     |
|                             |                             |            |                                            |                     |

\*\*\*\*\*